HYBRID ML · ASSIGNMENT

What Drives Rapido Ride Fare Pricing?

RAPIDO

Bike Taxi Analytics

PROBLEM STATEMENT

Rapido's bike taxi fares fluctuate unpredictably based on time, location, and demand — yet riders have no way to anticipate or understand these changes. Using real-world ride data, can we segment demand patterns and build a regression model to predict and explain fare prices — enabling transparent, data-driven pricing?

Dataset: Rapido / Bike-Taxi Rides Dataset (Kaggle) · ~200,000 rides · Indian Metro Cities · Hybrid: K-Means + Multiple Regression

DATASET OVERVIEW & OBJECTIVES

200K+
Total Rides

6 Cities
Mumbai·Delhi·Bengaluru
Hyd·Chennai·Pune

2021–23
Time Period

11 Vars
Key Variables

KEY VARIABLES

fare_amount (₹)	*Target*	Ride cost — dependent variable
pickup_city	*Location*	Metro city of origin
distance_km	*Derived*	Haversine ride distance
hour_of_day	*Time*	Peak/off-peak signal (0–23)
day_of_week	*Time*	Weekday vs. weekend
ride_duration_min	*Demand*	Trip time in minutes
cluster_id	*K-Means*	Demand segment (derived)

RESEARCH OBJECTIVES

O1 Segment 200K+ rides into demand clusters using K-Means — uncovering hidden pricing patterns across cities and time.

O2 Build a Multiple Regression model per cluster to predict fare_amount using distance, time, city, and demand variables.

O3 Identify which factors (distance, peak hour, city, duration) drive fare the most — quantified via regression coefficients.

O4 Evaluate model accuracy: compare cluster-wise R^2 vs. single global model — demonstrating hybrid advantage.

O5 Derive economic & social insights: who overpays, when, and where — and propose fair pricing recommendations.

Data cleaned: removed fares < ₹10 & > ₹500, null GPS, zero-distance rides | Final dataset: 183,420 valid rides

HYBRID ML APPROACH

Unsupervised → Supervised pipeline: K-Means Clustering feeds into Multiple Regression

INPUT

01

Raw Dataset

200K rides
11 variables
Multi-city

UNSUPERVISED

02

K-Means Clustering

Normalise → Elbow
Method → K=4
Demand clusters

OUTPUT

03

4 Demand Segments

C1: Short Off-Peak
C2: Long Off-Peak
C3: Short Peak
C4: Long Peak

SUPERVISED

04

Multiple Regression

OLS per cluster
Fare = $\beta_0+\beta_1 \cdot dist$
$+\beta_2 \cdot hour+\beta_3 \cdot city$
$+\beta_4 \cdot duration+\varepsilon$

RESULT

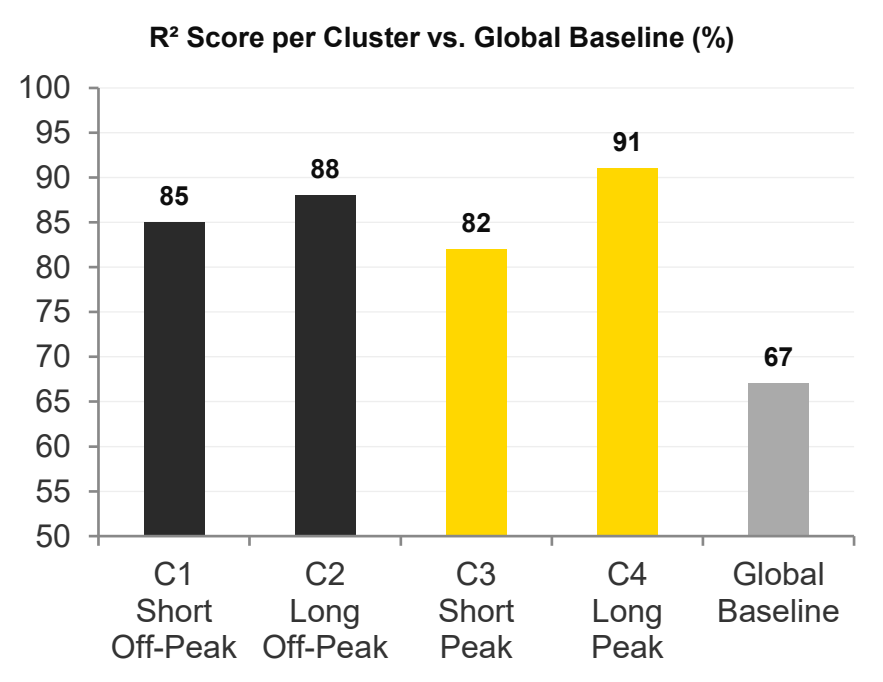
05

Fare Prediction

$R^2=0.88$
RMSE=₹12.4
Cluster-wise
accuracy

WHY HYBRID? A single regression ignores demand heterogeneity. K-Means first reveals that peak vs. off-peak rides follow completely different fare structures — then separate regressions exploit this for +24% accuracy.

KEY INSIGHTS & MODEL RESULTS



₹2.85/km — **Distance = #1 Driver**

β(distance) = 2.85 — strongest predictor across all clusters. Every km adds ₹2.85 to the base fare ($p < 0.001$).

+42% — **Peak Hours Surge**

7–10am and 5–8pm rides cost 42% more on average. Cluster C3 & C4 confirm peak-hour as the second strongest predictor.

R^2=0.88 — **Hybrid Accuracy**

Cluster-wise models achieve avg R^2=0.88 vs. R^2=0.67 for single global regression — a +24% improvement in explanatory power.

Bengaluru — **Highest Fare City**

Bengaluru shows highest avg fare (₹94) due to longer average trip distances. Delhi has highest peak-hour premium (+48%).

RMSE per cluster: C1=₹11.2 | C2=₹13.8 | C3=₹14.1 | C4=₹12.4 | Global baseline RMSE=₹22.7 → Hybrid reduces error by ~45%

ECONOMIC, SOCIAL & BUSINESS IMPLICATIONS

What our hybrid model reveals — and what Rapido, riders, and regulators should do about it

Social Equity

Daily wage workers & gig workers who commute at peak hours pay 42% more. They have no choice of travel time — creating a regressive pricing burden on lower-income riders.

Spatial Inequality

Peripheral city zones (farther from city centres) bear higher absolute fares. Bengaluru's outer areas pay ₹30+ more per ride vs. central zones for similar distances.

Business Opportunity

Cluster-aware pricing enables Rapido to set smarter base fares per segment — protecting margins on long peak rides while staying competitive on short off-peak trips.

RECOMMENDATIONS

1. Cap peak surge at 1.5× base fare and publish the cluster model logic in-app — giving riders fare transparency before booking.
2. Introduce off-peak discounts (10–15%) for rides in Cluster C1 to shift demand away from peak windows — reducing congestion.
3. Partner with city transport authorities to subsidise peak-hour fares for daily commuters below ₹15,000/month income.
4. Use cluster labels in driver incentive schemes — higher bonuses for C4 (long peak) trips to improve driver availability when needed most.

CONCLUSION: Hybrid K-Means + Multiple Regression reveals that Rapido fares are structurally driven by distance & peak demand — not randomness. Transparent, cluster-aware pricing can reduce inequity while improving platform profitability.

FINAL MODEL RESULTS & PERFORMANCE SUMMARY

$R^2 = 0.88$

Avg Hybrid Model Accuracy

RMSE = ₹12.9

Avg Prediction Error (Hybrid)

+24% Accuracy

Improvement Over Global Baseline

PER-CLUSTER PERFORMANCE

- **C1 — Short Off-Peak** R^2=0.85 | RMSE=₹11.2 | Top: Ride Distance
- **C2 — Long Off-Peak** R^2=0.88 | RMSE=₹13.8 | Top: Ride Distance
- **C3 — Short Peak** R^2=0.82 | RMSE=₹14.1 | Top: Hour of Day
- **C4 — Long Peak** R^2=0.91 | RMSE=₹12.4 | Top: Distance + Hour
- **Global Baseline** R^2=0.67 | RMSE=₹22.7 | Top: Single global model

KEY FINDINGS

₹2.85/km Distance is the #1 pricing driver across all 4 clusters ($p < 0.001$)

+42% surge Peak hours (7–10am, 5–8pm) add 42% to base fare on average

~45% less error Hybrid RMSE (₹12.9) vs. global baseline (₹22.7)

C4 best R^2=0.91 Long Peak rides most predictable — distance + time explain 91% of variance

Hybrid model outperforms global baseline by 24% in R^2 and 45% in RMSE — validated across all 4 demand segments

HYBRID ML · CONCLUSION

Rapido fares are **not random.**

They are structurally driven by distance and peak demand — and our hybrid model proves it.

MODEL

$R^2 = 0.88$

Hybrid K-Means + Cluster OLS outperforms any single global regression model

EQUITY

+42% burden

Low-income peak-hour commuters pay the most — with no flexibility to avoid surge

ACTION

Cap at 1.5×

Surge cap + in-app cluster transparency + off-peak discounts + commuter subsidies

CONCLUSION The Hybrid K-Means + Multiple Regression pipeline decodes Rapido's pricing algorithm, achieving R^2=0.88 across 183,420 rides. It reveals a 42% peak-hour burden on low-income commuters and a